T'Shara Haynes

Software Engineer | Research | Development

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WORK EXPERIENCE

Central Bank of Barbados - Bridgetown, Barbados
Governor's Office Intern **June 2025- August 2025**

- Mapped end-to-end payment system workflows using system documentation to support integration for the BiMPay rollout.
- Authored a research paper on fraud detection in instant payment systems, presented at the Annual Review Seminar.
- Produced clear process documentation to enable cross-team implementation.
- Reviewed regional data protection standards to ensure applicability of compliant use of AI and customer data.
- Developed fraud detection models to analyse transaction data and strengthen risk monitoring.

Pellucid IT Inc - Remote, Barbados
Project Assistant **October 2024- May 2025**

- Documented team workflows and improved task tracking to keep projects organized and on schedule.
- Implemented Scrum Methodology to enhance project agility and streamline workflows.
- Tracked timelines, identified blockers, and kept stakeholders informed to ensure smooth project delivery.

Headstarter AI - Remote, Trinidad and Tobago
Software Engineering Fellow **July 2024- September 2024**

- Created an inventory tracking application with React and Next.js to manage household supplies
- Developed a responsive portfolio website using HTML, CSS, and JavaScript to showcase technical projects
- Integrated OpenAI API with Next.JS to generate AI-powered study flashcards.

EDUCATION

University of the West Indies, Cave Hill Campus
Bachelor of Science in Computer Science w/ Environmental Science **Expected Graduation: May 2026**

SKILLS & INTERESTS

Programming Languages - Python, SQL, Java, PHP, JavaScript, HTML/CSS
Databases: MySQL, MongoDB
Frameworks/ Tools – NodeJS, Git, React, Streamlit, Bash, Pandas
Design Tools – Canva, Figma, Lightroom, Gimp, Aseprite
Methods– Technical Documentation, Agile/Scrum

PROJECTS

Fraud Detection Simulator **Research Project**
Python | Streamlit | SciKit-Learn

- Using Python and Streamlit, built a real-time transaction simulation app to model IPS payment flows.
- Applied Scikit-learn feature engineering and model evaluation techniques to detect anomalies in transaction data (~92% accuracy)
- Built Python-based simulation pipeline to ingest and analyse streaming transaction data in real-time
- Measured classification accuracy, precision-recall, ROC curves, and average detection time for each approach.
- Focused on challenges in low-label Caribbean IPS environments.

Energy Dashboard **Personal Project**
Python| Streamlit

- Developed a multi-building energy consumption dashboard with interactive visualisations, KPI metrics and scenari o simulation for usage reduction modelling.
- Implemented time-series forecasting using Facebook Prophet with custom date filtering and CSV functionality.

Sign Language Text-to-Speech Detector **Research Project**

Python| Tkinter | OpenCV

- Trained a machine learning model using Google's Teachable machine to classify sign language gestures.
- Pre-processed and labelled dataset; performed feature extraction to improve accuracy.
- Implemented real-time interference pipeline using OpenCV and Python.
- Designed and built an interactive Tkinter GUI that displayed signs as text and speech output.

RESEARCH & PUBLICATIONS

- Fraud Detection in Instant Payment Systems: A Caribbean Case Study of Machine Learning Constraints and Opportunities (2025).
- Augmentation of Machine Learning Algorithms for Air Quality Predictions (2026)
- Artificial Intelligence in Sign Language Text-to-Speech (2025)

LEADERSHIP

UWI Debate Society | University of the West Indies Cave Hill, Bridgetown, Barbados
Secretary **June 2024 - October 2025**

- Organised meetings facilitated communication between members and assisted in event planning and execution.
- Created an "Intro to Debate" booklet to engage with members and provide debating information to upcoming members and executive
- Increased engagement across social media pages by 13%

Faculty of Science and Technology Representative | University of the West Indies Cave Hill, Bridgetown, Barbados *Computer Science Representative* **June 2024 – May 2025**

- Acted as a liaison between students and faculty, addressed concerns, and coordinated academic and extracurricular initiatives for Computer Science students

Trinidad and Tobago Youth Advocacy Network| ***Trinidad and Tobago***
Environmental Department Member **October 2024 – Present**

- Engaged in advocacy initiatives focused on youth empowerment, policy change, and social impact.
- Collaborated with like-minded individuals and organizations to promote sustainable development goals (SDGs).

ADDITIONAL CERTIFICATES

AWS

- Getting Started with Security
- Introduction to Cloud 101

Coursera

- DevOps for Network Automation (NetDevOps)
- Google Project Management

REFERENCES

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